# Personal Investment Workbench Project
# Mid-Course Progress Report

**Student:** Menghua Wang (MenghuaW) — individual project, no team

**Course:** CSIS 4495, Fall 2026

**Report date:** October 18, 2026

**GitHub repository:** https://github.com/moonfa/F26_4495_S1_MenghuaW_C (work developed on branch feature/prompt-v2 and merged into main by pull request #1 on October 17, 2026)

**Demo video (public link):** https://youtu.be/YLuMH2WWGAM

**Product in one paragraph:** The Personal Investment Workbench is a local web application for one investor. It keeps a structured, versioned investment thesis for each company, tracks the portfolio and the reason behind every trade, and includes a separate valuation lab. The design rule is that code computes every number and decides what changed; the AI model (Google Gemini) only interprets and writes. Private data (holdings, plans, trade reasons, journal) is never sent to the model.

## 1. Work Date/Hours Logs

Phase I: work with ChatGPT and Gemini to clearify and test idea, output 1st version of project

Phase II: work with Claude to start over again and generate a more delicate project

### 1.1 Concept design (ChatGPT & Gemini)

### Workbench v1: version history

Each delivered version of the investment workbench with a summary of what it added or fixed. Dates are the local dates the versions were delivered.

**Note:** Dates are based on our project discussions and the development artifacts we worked on. Some Step 8.x changes happened iteratively within the same development period, so I use the most specific date/phase available rather than inventing exact timestamps.

Investment Research Workbench — Development & Discussion History

| Date / Phase | Version | Main Topic / Problem | Functions Added or Modified | Key Decisions / Outcome |
|---|---|---|---|---|
| **Sep 21, 2026** | **Initial Prototype** | Define the overall product concept and UI direction. | Prototype included **Dashboard, Company Research, Event Review, and Thesis History**. | The application should be an **investment research workspace**, not another Bloomberg/Yahoo Finance clone. External market-data services should provide raw data while the application focuses on research interpretation and decision workflow. |
| **Sep 22, 2026** | **Schema Design v1** | Define the core data model and the first implementation slice. | Introduced **Evidence Snapshot** as a first-class entity. Initial workflow: Ticker → Sync → OpenBB Adapter → Evidence Snapshot → Display. | Evidence must be stored as immutable snapshots so later AI analysis can be tied to the exact information available at that time. Thesis/version tracking would be added later. |

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| **Sep 25–26, 2026** | **Step 5 / AI Research Layer** | Add AI-generated investment research on top of collected evidence. | Added **Master Research Report**, **Key Takeaways**, and **Thread Actions**. Gemini analysis became evidence-aware and delta-aware. | AI should interpret evidence rather than simply summarize market data. Reports should follow causal reasoning such as **Observation → Mechanism → Financial Impact → What to Monitor**. |
| **Sep 25–27, 2026** | **Step 6 — Saved Research** | Preserve AI research instead of treating every analysis as disposable output. | Added saved historical reports and read-only loading. Reports could be reopened without calling OpenBB or Gemini again. | Historical reports became **immutable research anchors**. Loading saved research should not trigger external APIs. |
| **Sep 27, 2026** | **Step 6 — Research History** | Separate AI research from the investor's own judgment. | Added **Personal Research Journal** alongside historical AI reports. | Human notes should remain separate from AI-generated analysis. The system should preserve both without mixing authorship. |
| **Late Sep 2026** | **Step 6.1** | Improve historical report structure and valuation interpretation. | Extended ResearchReport with **valuation implications** and improved report metadata/history handling. | Valuation should remain an **implication/check layer**, not become the center of the research system. |
| **Late Sep 2026** | **Step 6 — Threads / Human Review** | Convert AI observations into persistent research topics. | Added ResearchThread, ThreadVersion, ThreadAction, and human review states such as **Approve / Edit / Reject**. | Long-term research state should not be changed automatically by AI. AI proposes changes; the human investor confirms them. |
| **Late Sep 2026** | **Step 8 — Evidence Update Architecture** | Full reports were too expensive and repetitive for every evidence refresh. | Added CompanyResearchState and ResearchUpdate. Introduced **Refresh Evidence**, **Analyze New Evidence**, and **Generate Full Review** as separate actions. | Established a three-level workflow: **Evidence Update → Research Update → Full Review**. A new snapshot should not automatically create a full report. |
| **Step 8** | **Evidence Classification** | Determine whether new evidence deserves AI analysis or a full review. | Added deterministic evidence_delta() and classify() logic based on metric changes and new news. | Routine evidence should be handled cheaply. AI should only be used when evidence reaches research significance. |
| **Step 8** | **Language Support** | Allow reports to be generated in English or Chinese. | Added report_language: en / zh-CN and report language persistence. | Language became a **generation-time property**. Changing the selector does not translate old immutable reports. This behavior was accepted as reasonable. |
| **Step 8** | **Watchlist / Research State** | Improve company management without deleting historical research. | Watchlist companies could move between **Watching / Following / Archived** states. | Removing a company from the active watchlist must **not delete historical evidence or reports**. |
| **Step 8** | **Timeline** | Provide a chronological view of research activity. | Added Research Timeline combining **Evidence Updates, Research Updates, Full Reviews, Thread Versions, and Personal Notes**. | Timeline represents **what happened**, while reports and threads represent research objects/state. |
| **Late Sep 2026** | **DB Compatibility Bug** | Historical reports disappeared and API requests returned HTTP 500 after a Step 8 installation. | Diagnosed SQLite error: no such column: research_reports.valuation_implications. | Root cause: SQLAlchemy create_all() creates missing tables but does **not ALTER existing tables**. Existing DB had an older schema. DB was repaired without deleting historical data. |
| **Step 8.1** | **UX Refinement** | Sidebar and Research page had excessive visual clutter. | Changed watchlist toward **ticker-only navigation**, improved historical report hierarchy, made Threads expandable/collapsible, and simplified timeline presentation. | Keep the application research-focused and compact rather than showing every internal object equally. |
| **Step 8.1** | **Classifier Guardrail** | Routine market/news changes could be incorrectly classified as material. | Reduced broad material keywords and introduced a conservative **materiality guardrail**. | A low-impact event should not recommend a Full Review merely because market data moved. |

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| **Step 8.1** | **Chinese Output Behavior** | Switching the language selector did not change existing reports. | Verified that newly generated content used the selected language while existing content remained unchanged. | Accepted as intentional: **historical reports remain immutable**; language selection controls future generation only. |
| **Step 8.1 Acceptance** | **WMT / MCD / MRVL Test Plan** | Need companies with different behavior to test classification. | Defined acceptance cases using **WMT, MCD, and MRVL**. | WMT = stable retail; MCD = mature franchise/consumer; MRVL = volatile semiconductor/AI. This gives a useful range for testing false positives and escalation. |
| **Step 8.1 Bug Fix** | **HTTP 500 — Analyze New Evidence** | MRVL POST /research-updates returned HTTP 500. | Traceback identified NameError: guardrail is not defined. guardrail(level, result) was added/restored. | Error was **not caused by excessive prompt size or Gemini reasoning**. It was a backend Python function-definition error. |
| **Step 8.2** | **Acceptance + Bug-Fix Pass** | Need systematic testing rather than manual UI-only testing. | Created acceptance and diagnostic scripts for WMT/MCD/MRVL. Tightened classifier logic and report-language cache behavior. | Acceptance testing should verify invariants such as history preservation, no automatic Full Review, and correct evidence/research separation. |
| **Step 8.2** | **Thread Action State Issue** | Previously approved/rejected Thread Actions sometimes appeared as pending again after a new report. | Investigated ThreadAction.review_status and report/action relationships. | Review status itself persists. The deeper issue is that a **new Master Report can generate a new ThreadAction row** representing a similar proposal. This should eventually be handled through semantic deduplication rather than blindly copying old approval status. |
| **Step 8.2 Discussion** | **Research Page Duplication** | Timeline, Historical Reports, Threads, and Thread Actions repeated similar information. | Defined semantic responsibilities for each object. | **Timeline = what happened; Historical Reports = complete research archives; Active Threads = what is currently monitored; Thread Actions = AI-proposed changes requiring human review.** |
| **Step 8.3** | **Research Workspace Consolidation** | Research page was becoming a collection of independent cards rather than one workflow. | Consolidated Current Research, Active Threads, Timeline, Archive, Journal, report viewer, and human-review flow. | Established the research flow: New Evidence → AI Interpretation → Thread Impact → Human Review → Persistent Research State → Historical Record. |
| **Step 8.3** | **Human Review UX** | Thread Actions and Active Threads visually duplicated each other. | Pending AI proposals were associated with affected Threads rather than treated as another equal research list. | **Active Threads represent approved persistent state. Thread Actions represent proposed changes.** They should not look like two competing lists. |
| **Step 8.3** | **Timeline Simplification** | Full Review entries repeated report summaries already available in Historical Reports. | Timeline Full Review entries were reduced toward event markers instead of reproducing report content. | Timeline should answer **"What happened?"**, not act as a second report library. |
| **Step 8.4 — First Attempt** | **Navigation Simplification** | Research page was still long and visually dense. | Experimented with Research sub-navigation: **Current / Master Report / History**. News renamed **Evidence**. Full Review/language moved under **More**. | Conceptually cleaner, but in actual use it fragmented the research workflow too much. |
| **Step 8.4** | **Overview Simplification** | Overview duplicated Financials and News/Evidence. | Preserved the existing primary Company Overview card and hid duplicated lower content. | Overview should eventually become a high-level dashboard with **multi-year and quarterly financial trends**, not duplicate detailed tabs. |
| **Step 8.4** | **Toolbar Simplification** | Too many primary buttons increased cognitive load. | Hid Load Saved Research; moved infrequent Full Review/language controls under **More**. | Primary workflow remains **Refresh Evidence → Analyze New Evidence**. Full Review should be exceptional. |
| **Step 8.4 Feedback** | **Research Navigation Regression** | Some tickers showed no report after Load Saved Research was | Investigation identified stale renderedReportId logic. | This was **not primarily a timestamp problem**. A report ID from a previously selected ticker could prevent the new ticker's latest report from loading. |

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| | | removed; History became difficult to use. | | |
| **Step 8.4 Feedback** | **Price / Header Design** | Price and 52-week range disappeared when viewing Financials/Research. | Decided to move ticker, company name, current price, and 52-week range into a **persistent company workspace header**. | Core market context should remain visible across Overview, Financials, Research, and Evidence. |
| **Step 8.4 Feedback** | **Company Description** | Long business description occupied too much vertical space. | Changed concept to a **collapsible, height-limited, scrollable business description**. | Preserve business context without dominating the page. |
| **Step 8.4 Feedback** | **Financial Metrics Semantics** | Asked whether Financial Health metrics correspond to current stock price and whether comparison is necessary. | Financial page concept separated into **Valuation Context**, **Financial Quality**, and **Analyst Consensus**. | Valuation multiples are price-sensitive; operating/TTM metrics are primarily fundamental; consensus is forward-looking. Future UI should add historical and peer comparison instead of treating a single number as sufficient. |
| **Step 8.4 Feedback** | **Price as Research Trigger** | Frequent price movements risk creating unnecessary reports. | Explicitly separated market-price movement from fundamental research significance conceptually. | **Price movement alone should not trigger a new Master Research Report.** Future logic should distinguish Market Change / Fundamental Change / Thesis Change. |
| **Step 8.4 Feedback** | **Report Impact Highlighting** | Asked whether changes to old research should use colors. | Proposed future semantic states: **Strengthened / Stable / Weakened / Needs Review**. | Deferred to Structured Thesis rather than inferring colors in the browser without real structured state. |
| **Step 8.4R** | **Revised Research Layout** | The Current / Master Report / History split felt fragmented and unnatural. | Removed the Step 8.4 Research sub-tabs. Returned to **one coherent Research page**. | Historical reports, Timeline, Personal Journal, Threads, current interpretation, and report viewer should remain logically connected rather than scattered across sub-pages. |
| **Step 8.4R** | **Historical Reports** | History showed timeline events but accessing full historical reports was awkward. | Restored **Historical Master Research Reports** as a visible list next to/with Timeline; selecting a report loads it into one report viewer. | Timeline and report archive stay together: one is the event index, the other is the complete research artifact. |
| **Step 8.4R** | **Automatic Latest Report Loading** | Some companies had saved reports but the report viewer appeared empty. | Changed logic to explicitly load the selected ticker's latest currentReportHistory[0]. | Fixes cross-ticker stale state from renderedReportId. |
| **Step 8.4R** | **Thread Quality Discussion** | Thread information remained too generic: unclear which financial metric or thesis it affected. | Determined that this cannot be solved properly with UI rearrangement alone. | Root issue is **schema + prompt structure**. Future Thread/Thesis output should explicitly identify affected thesis, financial metric, direction, mechanism, monitoring metric, and invalidation condition. |
| **Step 8.4R Installation Issue** | **Installer Reliability** | Step 8.4/8.4R installer ran but UI showed no change. User asked whether ux84.css/js needed to be copied manually. | Investigated packaging/install approach and created a **self-contained installer** embedding CSS/JS directly. Added SHA256 and marker verification. | CSS/JS files should **not need manual copying**. Installer should modify the actual app/static/index.html and verify that the running server is serving that file. |
| **Oct 8, 2026 — Current Direction** | **End of Step 8 / Toward Step 9** | Decide what should be stabilized before | Current architecture now centers on Evidence, Research Updates, Threads, | Next major improvement should be **Structured Thesis / Research State**, not another round of cosmetic UI patches. |

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| | | adding more AI features. | Human Review, immutable reports, and a simplified workspace. | |

### Functional evolution in one view

The project has effectively evolved through four architectural stages:

| Stage | Main Goal | Result |
|---|---|---|
| **Prototype / Evidence Layer** | Collect and preserve investment evidence | Ticker → Evidence Snapshot |
| **Step 5–6 Research Layer** | Turn evidence into durable AI research | Master Reports + Takeaways + Threads + Journal |
| **Step 8 Update Layer** | Avoid regenerating full reports for every change | Evidence Update → Research Update → Full Review |
| **Step 8.1–8.4R Workspace Layer** | Make the research system usable and remove duplication | Persistent Threads + Human Review + Timeline + Report Archive + simplified Research workspace |

After working with ChatGPT for 1 week, there are some problems seems unable to be solved, such as the UI is too ugly and insistance of data model. So I send all code to Claude and try to start over again to find a new way.

| Date | Hours | Work Completed |
|---|---|---|
| Sep 10, 2026 | 1 | Initial project idea and course requirement review. |
| Sep 11, 2026 | 1 | Refined the personal investment research use case. |
| Sep 12, 2026 | 3 | Compared existing finance platforms and project scope. |
| Sep 13, 2026 | 2 | Reviewed reusable financial-data sources and project boundaries. |
| Sep 14, 2026 | 1 | Tested company data needs using MRVL, MCD, UBER, and GOOGL. |
| Sep 15, 2026 | 1 | Shifted focus from data aggregation to personalized investment research. |
| Sep 16, 2026 | 1 | Designed the core Investment Thesis Engine workflow. |
| Sep 17, 2026 | 1 | Stress-tested the workflow across different company types. |
| Sep 18, 2026 | 1 | Reviewed and froze the core project design. |
| Sep 19, 2026 | 3 | Simplified proposal scope and aligned it with the course requirements. |
| Sep 22, 2026 | 5 | Research Information Architecture and Navigation Simplification |
| Sep 23, 2026 | 2 | Modify prompts for better analysis. Overview Tab Simplification |
| Sep 24, 2026 | 1 | Research Tab Restructuring: current, master report, history. |
| Sep 25, 2026 | 2 | Evolving News to Evidence, Handling Stock Price Changes |
| Sep 26, 2026 | 2 | Visualizing Impact and Changes |
| Sep 28, 2026 | 2 | Specific Implementation Notes |
| Sep 30, 2026 | 3 | Free Gemini token cannot support the model. Unable to improve further with ChatGPT, decide to redesign with support ot Claude |

## 1.2 Detailed design (Claude)

With support of Claude, the progress became much more smart and continually improved.

| Date | Hours | Work Completed |
|---|---|---|
| Sep 30, 2026 | 3 | Free Gemini token cannot support the model. Unable to improve further with ChatGPT, decide to redesign with support ot Claude |
| Oct 1, 2026 | 3 | Communicate with Claude for improvement on prompt, and better usage on token, smart and simple UI.<br>Reviewed the original repository (branch feature/step6-research-history) and listed improvements for analysis logic, token/prompt efficiency, pagination/timeline and UI |
| Oct 2, 2026 | 5 | Update architecture of porject. Better prompt and efficient token usage.<br>Designed v2 as a thesis tracker: Review rows as self-contained thesis states, deterministic change detection, no model call when nothing changed |

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| Date | Hours | Work Completed |
|---|---|---|
| Oct 3, 2026 | 3 | Analyse report is cut off and error. Communicate and solve the problem.<br>Fixed OpenBB start-up problems (pinned openbb 4.7.2; initialised OpenBB in the main thread) and set up the Gemini model fallback chain<br>Fixed truncated model output (thinking budget, higher output limits, retry once) and validated tickers before saving |
| Oct 4, 2026 | 5 | Update .env for multi-model switch when error.<br>Built the English front end: Brief, Thesis, Fundamentals, Timeline and Notes pages; grouped sidebar and Overview |
| Oct 5, 2026 | 2 | Design and test the personal investment portfolio part.<br>Position plans (buy zone, target, stop, rules) with code-evaluated status; trade reasons and post-trade review |
| Oct 6, 2026 | 5 | Add a Journal to record macro updates<br>Portfolio Journal with a code-generated review checklist; Overview period digest; tooltips explaining each field<br>Futu CSV import for holdings and trades (US / HK / SG symbols, de-duplication, ETF and cash types) and the Portfolio page<br>Prompt tracker-1.6: evidence discipline, code-computed implied valuation metrics, thesis health (Intact / Watch / At risk) |
| Oct 7, 2026 | 2 | Modify Journal page and add Chinese translation function<br>On-demand Chinese translation of stored analyses; share-class symbol fix (BRK.B to BRK-B) |
| Oct 8, 2026 | 5 | Update design report and fortify system security<br>Modify unclear display of HK stock name<br>Trade calendar design (month / week view, journal events layer)<br>Security hardening (Host / Origin checks, upload limit, headers, safe links) and the English design document |
| Oct 9, 2026 | 4 | Trade outcomes and guidance: the reason dialog defaults to the trade day; result rating (worked / mixed / failed) and review text per trade (v2.14.0). |
| Oct 10, 2026 | 3 | Position sizing in money (target and maximum position in the stock's own currency) with an over-maximum alert (v2.15.x). Started the valuation lab as a separate page and a separate append-only store: P/E and EV/EBITDA exit-multiple models with Bear / Base / Bull ranges (v2.18.0). |
| Oct 11, 2026 | 2 | Valuation lab: P/FCF method, Compare tab for 2 to 5 tickers ranked by the most conservative Base result, restyled to match the workbench, method-fit guidance and growth references (v2.19–2.21). |
| Oct 12, 2026 | 2 | Tested the valuation lab on different business types, including a turnaround case (DEO) where reported earnings are far below what analysts expect; prepared the code for an outside review. |
| Oct 13, 2026 | 2 | Valuation lab: start from the furthest analyst year with the exit multiple on the same basis (stops a recovery being counted twice); earnings-dislocation detection and a "what you must believe" grid; help on every Compare input; Valuation tab on the company page with out-of-date flags and links from thesis assumptions to valuation inputs (v2.23.x). Compared ChatGPT's review of the valuation design with the code and adopted only the valid points. |
| Oct 14, 2026 | 3 | Valuation records: replace unused imports, delete a saved record after confirmation; English / Chinese switch for the valuation lab; a question-to-write-down card on every company page; fixed stale tests; README rewritten for GitHub (v2.24.0). Checked the code in to GitHub. |
| Oct 16, 2026 | 3 | Companies quoted in one currency and reporting in another (0883.HK) are converted at one labelled spot rate; Fundamentals tab regrouped into what you pay / what you get / the record; Overview digest titles aligned with their sections and tiles that jump to them; sort by best or worst P&L (v2.25.0). |
| Oct 17, 2026 | 3 | Discussed adding Barron's and Seeking Alpha content and compared the answers of Claude and ChatGPT; tried analyst-forecast columns in Fundamentals and a reading-capture bookmark, then removed both (v2.26.0 withdrawn); valuation page returns to the same company (v2.27.0); reviewed an outside code review against the code; ran the full pytest suite (139 tests) and fixed the one stale test; merged the work into main on GitHub. |
| Oct 18, 2026 | 5 | Mid-course report, demo plan and README/installation check. |

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## Workbench v2: version history

| Version | Delivered | Features added or fixed |
|---|---|---|
| 2.5.1 | 10/03 | Foundation of v2: thesis tracker (each Review is a self-contained thesis state); deterministic change detection with price-effect stripping and a free 'check' when nothing changed; token strategy (compact JSON, thinking budget, retry on truncation); English front end (Brief, Thesis, Fundamentals, Timeline, Notes); grouped sidebar and Overview; Futu CSV import for holdings and trades (US / HK / SG); position plans with code-evaluated status; ticker validation; in-page confirmation dialogs; Gemini model fallback chain; prompt tracker-1.6 with evidence discipline and code-computed valuation metrics. |
| 2.6.0 | 10/05 | Trade reasons and review: strategy, fundamental and technical reasons, revisit date and outcome per trade; pre-split price flag; plan tags on Overview; Portfolio card listing trades that need a reason. |
| 2.7.0 | 10/06 | Portfolio Journal (decision / review / macro / source) with a portfolio snapshot saved on each entry; code-generated review checklist and a Start review prefill; journal entries and notes merged into the company timeline; trade-import diagnostic report (rows, unfilled, unreadable, symbols) to investigate missing TLT trades. |
| 2.8.0 | 10/07 | Search, filters, sorting and paging for Overview and Portfolio; sector column; allocation bar by sector, investment purpose, type and view; visual polish (row colour bars, highlighted summary cards); Analyse now asks for confirmation and shows progress; hover help bubbles on Journal fields; sidebar filter box. |
| 2.9.0 | 10/07 | On-demand Chinese translation stored per review (buttons for latest, historical and baseline text); market filter (US / HK / SG); help bubbles that define View, Thesis health, assumption status, plan status and type; collapsible sidebar groups with fixed header and footer; share-class symbols fixed (BRK.B to BRK-B) with automatic migration of old rows. |
| 2.9.1 | 10/07 | Fixes found in use: translate button for the older baseline report; caption explaining the status dots; help bubbles no longer clipped by scrolling tables; bond ETFs recognised by name keywords. |
| 2.10.0 | 10/07 | Overview Period digest (7 / 30 / 90 days): trades, thesis changes, plan alerts, due items, biggest movers by weight x move; computed by code with no model call. |
| 2.11.0 | 10/07 | Security hardening: Host and Origin checks against cross-site and DNS-rebinding requests, upload size limit, security response headers and a content security policy, http(s)-only links; company names for Hong Kong and Singapore codes in the sidebar; full English design document including a Security chapter. |
| 2.12.0 | 10/08 | Trade calendar: month, week and list views (Sunday start); chips showing buy / sell and review state; day detail with reason, review and change since the trade; round-trip and pre-split flags; filters; layers for due items and journal events; journal entries get an optional event date. |
| 2.13.0 | 10/08 | Calendar legend and help bubble; '+ Journal note or event' from any calendar day (event date prefilled); cross-links so Portfolio, Overview and Journal point to the calendar for trade detail; Journal checklist date picker; Portfolio backlog card reduced to a summary plus the 5 most recent trades. |
| 2.14.0 | 10/09 | Trade outcomes and guidance: reason dialog defaults the decision date to the trade day (editable); result rating (worked / mixed / did not work / too early) with what happened and follow-up; one reason for several selected trades; ✓ ◐ ✗ markers; outcome counts by strategy on Portfolio; in-app Guide with the decision loop, module map and the goal and standard of each holding strategy. |
| 2.15.0 | 10/10 | Position sizing and logic: Plan gains target and max weight (percent), an over-max-weight alert and an estimated amount to add or trim; the position logic and strategy are written once in the Plan and copied into trades, which now record only role and trigger; “use this logic” for trades without a reason; quarterly (8) and annual (5) results in Fundamentals; code-computed quarterly trend for the model (prompt tracker-1.7); a newly reported quarter counts as a change. |

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| 2.15.1 | 10/10 | Position size set in money, not percent: Plan has a target position and a maximum position in the stock's own currency, compared with quantity x price (no portfolio-value estimate needed); the plan card shows room left and an over-max-position alert; Portfolio keeps broker percentage weights; help-icon layout fixed in dialogs. |
| 2.18.0 | 10/10 | Valuation lab as a separate page and a separate append-only store (valuation.db): P/E and EV/EBITDA exit-multiple models with Bear / Base / Bull ranges and reverse calculations (what EPS, margin or multiple earns the required return); EPS anchors labelled by kind (trailing twelve months, analyst year, reported year); status-quo defaults; plausibility checks that never change the numbers; analysis records keep only a reference to their data record; page and server API version check. Includes 2.17.0, the first version of the lab. |
| 2.19.0 | 10/11 | P/FCF method (free cash flow per share, before or after stock-based compensation); Compare tab for 2 to 5 tickers on the same rules, ranked by the most conservative Base result, methods never averaged; saved comparisons. |
| 2.20.0 | 10/11 | Valuation page restyled to match the workbench (tabs, dark mode, help bubbles); growth fade to a long-run rate; exit-multiple references (today, own history, peers, sector range); optional industry peers from Yahoo; de-rating case in Compare. |
| 2.21.0 | 10/11 | Method-fit guidance (suits / use with care / not suitable, with a recommended method); growth references with Use buttons; default fade of 5% a year; share count from market cap ÷ price; enterprise-value reconciliation; implausible default results flagged; year-by-year Base path; analyst-versus-scenario note. |
| 2.23.0 | 10/13 | Valuation starts from the furthest analyst year and keeps the exit multiple on the same basis, which stops a recovery being counted twice (the DEO case); earnings-dislocation detection and a "what you must believe" grid; "?" help on every Compare input and column; review fixes (P/FCF multiple pairing, fractional-year growth, implausible rows kept but not ranked, de-rating reference from peers or own history only, share-count guard); Valuation tab on the company page with out-of-date flags, return at the Plan price, links from thesis assumptions to valuation inputs and a question saved to Notes; prompt tracker-1.8; README for the repository. |
| 2.23.1 | 10/13 | Conclusion shown as a highlighted block; the result states which fiscal year the exit EPS belongs to; help on Save snapshot; the saved-snapshot list shows method, Base return or price in local time; Guide section on what gets saved. |
| 2.24.0 | 10/14 | Valuation records: re-importing a ticker replaces earlier imports nothing uses, any saved record can be deleted after confirmation (data still used by a saved result is protected), and unused imports can be cleared; English / Chinese switch for the valuation lab (Guide, help bubbles, conclusion and notes); the question to write down is shown on the company page for every company, editable, with an optional look-again date; fixes to the lifecycle and API tests; README rewritten for GitHub with an app overview. |
| 2.25.0 | 10/16 | Other currencies (0883.HK: HKD price, CNY accounts) converted at one spot rate and labelled; Fundamentals tab regrouped into what you pay / what you get / the record; Overview digest titles match their sections, tiles jump to them; sort by best/worst P&L and weight. |
| 2.26.0 | 10/17 | Withdrawn. Analyst-consensus columns in the Fundamentals tables and a bookmark button for saving readings were tried and removed in 2.27.0. |
| 2.27.0 | 10/17 | Valuation lab "back" opens the same company on its Valuation tab (Overview if none). Final design version. |

No separate archive exists for 2.16.0 or 2.17.0 (the first valuation lab, described under 2.18.0), nor for 2.22.x, which was an external review copy and is not part of this line. Dates for 2.5.1 to 2.7.0 and 2.18.0 to 2.20.0 are the work-log dates of that stage; the repository holds fewer, larger check-ins than there were versions (see 2.5).

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## 2. Repo Check

Repository: https://github.com/moonfa/F26_4495_S1_MenghuaW_C.git

(branch: feature/prompt-v2, merged into main by pull request #1 on Oct 17, 2026)

### 2.1 Repository folder

| folder | File/sub-folder | Features |
|---|---|---|
| Implementation | -workbench_v2<br>-project_dump.txt | Project code: FastAPI backend, two single-page front ends (workbench and valuation lab), 139 automated tests, README with installation steps, docs/DESIGN.md. project_dump.txt is a text export of the earlier v1 code. |
| ReportsAndDocuments | -Proposal (final)<br>-Work&AI_Log<br>-ProgressReport1<br>-<br>Workbench_v2_Version_History<br>-this Mid-Course Report | Proposal; work log and AI prompt record; first progress report; version history of v2; this report. |
| Requirments | -4495_ProposalTemplate<br>-CSIS4495_IntroDocument | Course requirement documents (the folder name is spelled this way in the repository). |
| Misc | -diagnosis and optimisation plan<br>-v2 design document (Chinese) | Earlier design and diagnosis documents written during the v1 to v2 redesign. |

### 2.2 Architecture

```
OpenBB (yfinance) -> Snapshot -> signals.compute_delta / classify
                                   | none -> Review(kind=check)   free, deterministic
                                   | minor / material
                                   v
                  prompts + provider (Gemini) -> Review(baseline | update | rebaseline)
                                                  |
Broker CSV -> Holding / Trade --+                 |
Your input -> Plan / TradeReason / Note / JournalEntry
                          v                       v
                    FastAPI (/api/v1)  <---------+ -> frontend (index.html)

Valuation lab (no AI, separate file valuation.db, append-only):
Yahoo data -> import (data) -> P/E, P/FCF, EV/EBITDA -> Save snapshot (analysis) / Compare
(comparison)
                                                        -> read-only Valuation tab on the com
pany page
```

| Module | Responsibility |
|---|---|
| app/adapters/ | OpenBB adapter (openbb pinned to 4.7.2, yfinance provider); multi-route news fallback; valuation_adapter.py and valuation_references.py import prices, statements, analyst estimates and peers for the valuation lab |
| app/core/signals.py | Change detection, fingerprint, evidence basis, data gaps, code-computed derived valuation metrics (price-effect stripping) |
| app/core/thesis.py | Thesis health (Intact / Watch / At risk), sector KPI hints |
| app/core/plan.py | Position plan status (buy zone, target, stop, maximum position in money) |
| app/core/translate.py | Extraction and re-insertion of translatable text fields (on-demand Chinese) |
| app/core/valuation.py, valuation_context.py, multiples.py | Valuation arithmetic for P/E, P/FCF and EV/EBITDA (starting value x growth path x exit multiple), Bear / Base / Bull, reverse calculations, “what you must believe” grid, earnings-dislocation detection, method-fit |

| Module | Responsibility |
|---|---|
| app/core/compare.py, snapshots.py, fx.py | Compare ranking of 2 to 5 companies; list, delete and prune of saved valuation records; currency conversion when price and accounts differ |
| app/core/valuation_link.py, app/api/valuation_link.py | Read-only link between saved valuations and the company page: out-of-date flags, return at the Plan price, assumption-to-valuation-input links |
| app/ai/ | Schemas, prompts (version tracker-1.8), provider with a Gemini model fallback chain |
| app/portfolio.py | Broker CSV parsing (holdings, trades) and symbol normalisation (US / HK / SG) |
| app/service.py | Refresh orchestration: validation, caching, choosing the analysis type, calling the model |
| app/security.py | Host / Origin checks, upload limit, response headers |
| app/core/calendar.py | Market-local trade dates, quick round-trip detection |
| app/api/routes.py, journal.py, calendar.py, valuation.py | HTTP endpoints |
| app/static/index.html, valuation.html | Two front ends in plain JavaScript: the workbench and the valuation lab (English, with an on-demand Chinese layer) |
| tests/ | pytest unit and API tests (139) and a front-end smoke test with a simulated DOM (node tests/ui_smoke.js) |

Technology: FastAPI, SQLAlchemy 2 on SQLite, pydantic v2, OpenBB 4.7.2, Google Gemini through the google-genai SDK, vanilla JavaScript.

## 2.3 Data Model

| Table | Purpose |
|---|---|
| companies | Symbol, name, group (holding / watching / idea / archived) |
| snapshots | Normalised evidence for each refresh, plus price and fingerprint |
| reviews | The core table. Each row is a complete, self-contained thesis state: kind (baseline / rebaseline / update / check), level, view, view_change, one_liner, state, changes, narrative, delta, usage, and an optional Chinese translation zh. |
| notes | Per-company notes: decision log, comment, source, question. A note can attach to one assumption, and stores the price at the time and a revisit date. |
| holdings | Broker holdings snapshot: quantity, cost, P&L, weight, currency, type (stock / etf / cash) |
| trades | Filled orders, de-duplicated by a content hash |
| plans | Position plan: style, buy zone, target price, stop price, add / trim / exit rules |
| trade_reasons | Why each trade was made (strategy, fundamental and technical reasons, revisit date) and the later outcome review |
| journal_entries | Portfolio journal: type, title, body, related symbols, revisit date, optional event date (for market or world events), and a snapshot of the portfolio when written |
| assumption_drivers | The user's own link from a thesis assumption to the valuation input it depends on (growth, margin, exit multiple, share count, net debt) |
| valuation.db (separate SQLite file) | Valuation lab store: one table valuations(id, ticker, kind = data \| analysis \| comparison, created_at, payload). Append-only: a record is never edited; the user may delete one, and imported data still used by a saved result is protected |

Because every review is self-contained, the timeline is simply the list of reviews, and a version comparison is the difference between two `state` values. Reviews are append-only: an analysis never overwrites an earlier one.

At start-up `ensure_columns()` adds any columns that a newer model has but an old SQLite table lacks, and `normalize_symbols()` rewrites old records such as `BRK.B` to `BRK-B`.

## 2.4 Change to Proposal

Added functions: investment portfolio management (broker CSV import, position plans, trade reasons and outcome review), journal, notes, trade calendar, on-demand Chinese translation, security hardening.

Companies analysed: the plan changed from 4 companies to about 60 (33 holdings plus watch-list and idea companies).

Added a valuation lab (P/E, P/FCF and EV/EBITDA exit-multiple models, Compare, link to the company page). The proposal had limited valuation to a light check; it grew because comparing what the price implies across companies turned out to be the most useful way to question a thesis.

Reduced: no AI is used on the valuation page, and no back-test or prediction score is claimed.

## 2.5 Check-in history

Commits to the v2 repository (git author dates). The earlier v1 repository, moonfa/F26_4495_S1_MenghuaW, holds 14 commits from September 19 to 30.

| Date (2026) | Commit message | Contents |
|---|---|---|
| Oct 3, 17:42 | first commit | Repository for v2 created. |
| Oct 3, 18:01 | update version by Claude | v2 foundation: thesis tracker, change detection, OpenBB and Gemini fixes, tests; proposal and requirement documents. |
| Oct 4, 21:05 | Evidence basis, data gaps, derived price metrics; SGD import | Evidence basis and data-gap lines, derived valuation metrics, Singapore symbols in the broker import. |
| Oct 5, 22:14 | update gemini models and add portfolio, plan | Gemini model fallback chain, portfolio import, position plans, trade reasons. |
| Oct 5, 23:13 | add work log and ai prompts record file | Work & AI log document. |
| Oct 8, 00:16 | update journal, translation and security | Journal, Chinese translation, trade calendar, security hardening, design document. |
| Oct 14, 12:11 | update with valuation models and fixed bug | Valuation lab (through v2.24.0), progress report 1. |
| Oct 14, 12:15 | Initialize README for Personal Investment Workbench v2 | README. |
| Oct 17, 17:59 | add valuation model and modify UI | Valuation lab updates through v2.27.0, version history. |
| Oct 17, 18:03 / 18:07 | Merge ... resolve README conflicts / Merge pull request #1 | feature/prompt-v2 merged into main. |

The repository holds 11 commits over 15 days for 27 delivered versions, so several versions were checked in together (for example v2.18 to v2.24 on October 14). Work is checked in at the end of each working stage rather than after each version.

### 2.6 Installation: how to run the demo

1. Requirements: Python 3.10 or newer (tested on python 3.13 no Gemini key is needed for a demo.
2. git clone https://github.com/moonfa/F26_4495_S1_MenghuaW_C.git, then cd Implementation/workbench_v2.
3. Create and activate a virtual environment, then pip install -r requirements.txt (installs OpenBB 4.7.2).
4. Copy .env.example to .env and set AI_PROVIDER=mock (placeholder analysis, no key) or AI_PROVIDER=gemini with GEMINI_API_KEY and AI_MODEL.
5. python -m uvicorn app.main:app --reload --reload-dir app, then open http://127.0.0.1:8000/. The valuation lab is at /valuation.
6. Type a ticker (for example MSFT) in Add ticker. Portfolio, Journal and Calendar work after importing a broker CSV from the Portfolio page. The valuation lab needs no key at all.
7. Tests: pip install -r requirements-dev.txt, then python -m pytest -q (139 tests) and node tests/ui_smoke.js.

The full steps, the settings table and troubleshooting are in the repository README (Implementation/workbench_v2/README.md).

## 3. Summary of Work Completed

### 3.1 What works now

| Area | What works now | Main code |
|---|---|---|
| Company research | Add a ticker, pull market data through OpenBB, and keep a structured thesis (assumptions with status, kill criteria, monitor list). Brief, Thesis, Fundamentals, Timeline and Notes pages per company. | ai/, core/signals.py, thesis.py, service.py |
| Change detection | Each refresh is compared with the last analysed state. The price effect is removed from multiples, so a rise in P/E that only follows the price is not a change. If nothing changed, a free check row is saved and no model call is made. | core/signals.py |
| Portfolio and plans | Broker CSV import for holdings and trades (US, HK, SG), allocation views, position plans (buy zone, target, stop, maximum position in money) with code-evaluated alerts. | portfolio.py, core/plan.py |
| Trades, journal, calendar | A reason and later outcome review for every trade; portfolio journal with a code-built review checklist; month, week and list calendar; period digest on the Overview with sort by P&L and weight. | api/journal.py, calendar.py |
| Valuation lab | Three exit-multiple methods, Bear / Base / Bull, reverse calculation, “what you must believe” grid, earnings-dislocation detection, Compare of 2 to 5 companies, other-currency conversion, saved snapshots that can be listed and deleted, English / Chinese switch. No AI is used. | core/valuation*.py, compare.py, fx.py, valuation.html |
| Link to the company page | The Valuation tab shows what the saved assumptions imply today and at the Plan price, flags a saved valuation as out of date, ties thesis assumptions to valuation inputs and saves a question to Notes. It changes no plan, thesis or saved valuation. | core/valuation_link.py |
| Language and privacy | Analyses are written in English; Chinese is produced on demand and stored. Holdings, plans, trade reasons and the | core/translate.py, security.py |

| | | |
|---|---|---|
| | journal never go to the model. Local-only server with Host / Origin checks and upload limits. | |

## 3.2 Technical highlights

• Code computes, AI interprets. Every number, every "what changed" decision and every plan or valuation result is calculated in tested Python; the model only writes the interpretation. This avoids invented figures.

• No change, no tokens. A fingerprint of the evidence (without the price) and an anchor to the last analysed state decide whether a model call is needed at all; unchanged refreshes are free.

• Self-contained, append-only reviews. Each review row is a complete thesis state, so the timeline is the list of reviews and a version comparison is the difference of two states; nothing is overwritten.

• Valuation on one basis. The starting value (trailing or an analyst year) and the exit multiple are on the same basis, and growth is counted for exactly the holding period. This removed a double count when a recovery was priced in twice (the DEO case).

• Honest outputs. Plausibility checks never change the numbers; results that cannot be compared are shown but not ranked; the limits (no back-test, dividends excluded, a single free data source) are stated on the page and in the design document.

• Privacy boundary. Private data is separated from what the model receives, and the valuation page makes no model call.

## 3.3 Size and quality checks

About 5,600 lines of Python in app/, two single-file front ends, and about 1,550 lines of tests. The pytest suite has 139 tests (all pass on the author's machine; the run found one outdated test, which was corrected) and a front-end smoke test runs 83 checks on a simulated page. The design document (docs/DESIGN.md, about 500 lines) records the decisions, rejected options and known limits.

## 3.4 Known limits

• No back-test and no prediction track record; Bear / Base / Bull are the investor's ranges, not probabilities.

• Price return excludes dividends, tax and fees; the three valuation methods are not independent of each other.

• One free data source (Yahoo Finance through OpenBB); figures are provider fields, not reports.

• Single local user, no login; the Chinese layer does not cover table headings or notes written by the backend.

• Small improvements noted for later: prune old evidence snapshots, show a loss-to-profit swing as such instead of a very large percentage, and split routes.py.

## 3.5 Remaining work

• Use the workbench on the real portfolio for a period and record what proves useful or not.

• Apply the small improvements above if the testing shows they matter.

• Final report, demonstration and a clean repository (documents and code only, no local files).

### 3.6 In my own words

[Write 3 to 5 sentences in your own words: what you accomplished so far, what was hardest, and what the project adds beyond using ChatGPT or Yahoo Finance directly. This is the part of the report that has to be your own voice.]

## 4. AI Use Section

### 4.1 AI Tools used

| AI Tool Name | Version, Account Type | Specific feature for which the AI tool was used | Value Addition |
|---|---|---|---|
| Claude | Claude Sonnet 5.5; free plan | Reviewing the original project; generating and refactoring the v2 backend and the front ends (workbench and valuation lab); writing unit tests, API tests and a simulated-DOM UI test; drafting the design document, README and version history; code and design discussions on valuation, security and scope. | Very logic, functional, considerate<br>Need to wait several hours for reset |
| Google Gemini API (inside the app) | AI_PROVIDER=gemini<br>AI_MODEL=gemini-3.8-flash<br>AI_MODEL_FALLBACKS=gemini-3.7-flash,gemini-3.6-flash,gemini-3.5-flash,gemini-3-flash,gemini-3.5-flash-lite,gemini-3.1-flash-lite | Runtime analysis engine: produces structured thesis baselines and updates, and Chinese translations of stored analyses (not used to write the report) | Free token<br>Multi model switch, solid response |
| ChatGPT | GPT-5.6 Sol plus | Reviewing the original project; generating and refactoring the v1 backend and single-file front end; writing unit tests; drafting the design document. In October also used as a second opinion on the v2 valuation design and on Claude's answers (see the cross-check table). | ChatGPT plus is not as logic and well designed as Claude |
| Muse | [enter version and account type] | Outside code review of the whole project (Oct 17); each point was checked against the code before deciding whether to act. | [enter in your own words] |

How AI output was checked. Output from one tool was not accepted on trust; it was compared with the code, with the tests, or with another tool, and only the points that held up were used.

| Date (2026) | What was compared | Outcome |
|---|---|---|
| Oct 13 | ChatGPT review of the valuation design, compared with Claude's code | Adopted: P/FCF multiple pairing, fractional-year growth, implausible rows kept but not ranked, de-rating reference only from peers or own history, share-count guard. Not adopted: trailing earnings as the default start (a turnaround would be mispriced) and a sum-of-the-parts method (needs data the free source does not give). |
| Oct 17 | Claude and ChatGPT on adding Barron's and Seeking Alpha | Both agreed not to scrape paid or restricted content and not to let outside opinions change a thesis. A save-a-link bookmark was built, tried, found unnecessary and removed. |
| Oct 17 | Muse code review (7 points) | Each point was checked in the code: 3 held (evidence snapshots grow without pruning; two configuration sources; routes.py is long), 4 did not (the refresh request does not block other requests; assumption ids are validated and generated by code; the Review language default is overwritten; the OpenBB adapter constructor is trivial). No code change was needed before submission. |

| | | |
|---|---|---|
| Oct 18 | Full pytest run on the author’s machine | 139 tests: 138 passed, one outdated test was found and corrected (it looked for old error wording after currency conversion was added). |

## 4.2 AI Prompt history (ChatGPT & Gemini, Phase I)

### Prompt history (summary)

The following is a concise record of the major prompts used during project formation. The complete prompt history will be retained with the project documentation.

| Stage | Prompt / Request | Decision |
|---|---|---|
| Initial idea | "For this course's project, I plan to design a platform tailored for a self-employed financial content creator. Please help me continuously refine and iterate on the concept and proposal for this project through a series of Q&As and discussions. Here is my initial outline for the project:<br><br>A platform for financial creators to generate and publish content, as well as manage followers;<br><br>Automated aggregation of daily market and financial information into daily/weekly posts. The generated content should automatically interface with feeds on platforms like X, WeChat, and Xueqiu, and be published after my manual review and approval;<br><br>Utilizing tools like AI agents to automate workflows;<br><br>Benchmarking against existing self-media platform models used by financial content creators;<br><br>Optional feature: Employing AI agents to track and simulate the perspectives of various financial creators, host discussions on specific topics, and generate preliminary view summaries;<br><br>Managing followers and subscriptions across different communities and platforms;<br><br>Aggregating member comments and replies from various platforms, auto-liking their responses, categorizing their questions, generating reply drafts, and publishing them upon my confirmation." | Cut scope:<br>- The generated content should automatically interface with feeds on platforms like X, WeChat, and Xueqiu;<br>- Optional feature: Employing AI agents to track and simulate the perspectives of various financial creators, host discussions on specific topics, and generate preliminary view summaries;<br>- Managing followers and subscriptions across different communities and platforms;<br>- Aggregating member comments and replies from various platforms, auto-liking their responses, categorizing their questions, generating reply drafts, and publishing them upon my confirmation." |

# Mid-Course Progress Report — Menghua Wang

| Stage | Prompt / Request | Decision |
|---|---|---|
| | I cannot answer these questions definitively just yet. I would like you to provide more background regarding the workload and complexity involved so that I can clarify the target positioning of this project.<br><br>There are already many financial content creators in the market tracking and sharing information. My platform should focus on comprehensive coverage of key market information alongside continuous tracking of targeted focus companies and industries;<br><br>As an individual investor, I want to build this platform to build up my own research framework, acquire information quickly, and improve my investment performance through output-oriented practice, while accumulating a community of like-minded members;<br><br>This process should be progressive, moving from simpler to more complex phases. For the current project scope, what core features should be prioritized in Phase 1 of the platform design?<br><br>Additionally, what would be the cost, maintenance overhead, and operational difficulty associated with this?" | |
| Scope check | Many finance websites already provide company data; what should the project add? | Do not rebuild a financial terminal. |
| | A follow-up question: When building the scanner/tracker for macroeconomic, industry, and company-specific information, should I directly leverage data and feeds from existing platforms? Or are there suitable open-source repositories/codebases available? Or should I design and develop it from scratch based on certain standards? Which existing tools, APIs, or resources can be referenced, cited, or directly integrated? | Data source: yfinance, futu. OpenBB |
| | This is a solo project that I will complete entirely on my own. The primary purpose of the platform is to serve my own research productivity, help me gain experience, and systematically output content for publishing. Therefore, it is critical to keep the project scope strictly under control. Can the current workload realistically be managed by a single person? Furthermore, what platform or technology stack should I build this on? Power Apps? Or something else? I noticed some creators use Patreon, others use Knowledge Planet, Discord, or GitHub—are those the kinds of platforms I should be leveraging? | IT architecture |
| | Valuation models are quite complex: first, there are many valuation methodologies; second, different types of companies require different approaches (e.g., cyclical vs. non-cyclical stocks, sum-of-the-parts vs. direct consolidation, or selecting appropriate P/E multiples); third, while existing financial websites provide earnings and growth forecasts, accessing them often requires paid subscriptions. Are there free data sources available? How can I build and maintain this valuation model in a low-cost, sustainable way? | No valuation model design |
| Data model | Compare MRVL, MCD, UBER, and GOOGL company information needs. | Identify reusable and company-specific research elements. |
| | I feel that manually extracting data to build company profile pages might result in an interface inferior to Yahoo Finance, leading to redundant work with limited practical returns. Let's continue discussing how to pivot the project design to maximize ROI and align better with my goals. For the free content available on Yahoo Finance, can I directly capture/embed the charts, or should I simply link out to them if embedding isn't feasible? This would allow us to shift the design focus toward the custom and advanced features I truly need. | No duplicated work similar to yfinance, and focus on value added engine. |

| Stage | Prompt / Request | Decision |
|---|---|---|
| Core engine | Design a workflow from personal thesis to new evidence and thesis update. | Investment Thesis Engine becomes the core analysis workflow. |
| | What are the main technical hurdles and highlights of this project? Could the core focus be valuation models, stock scoring, or business classification? Should these models be built custom from scratch, or generated by an AI agent? Or should the AI be tasked with scraping data from relevant financial websites and research reports? | Design investment thesis engine |
| | How should I handle discrepancies between external market valuations, event impact analyses, and my own target valuation? Do I need to list all of them?<br>(Re-confirming): Yes, I believe your assessment is correct, and we should adjust the project design accordingly based on this approach. | The system will not implement an advanced or universal valuation engine. Instead, it will support lightweight valuation review by storing selected personal assumptions and comparing them with a small set of external market-reference metrics from existing financial platforms. |
| Stress test | Apply the workflow to four different business models. | Add flexible Sub-Thesis, Driver, Risk, KPI, and Event Impact concepts. |
| Proposal review | Reduce scope for a 120-hour solo course project. | Use one primary case, small validation set, simple evaluation, and limited valuation support. |
| Workbench v1 design | V1 is developed and modified more than 20 round with the support of ChatGPT. | |
| Workbench v2 design | V2 is updated on the V1.<br>API, analysis logic and UI are redesigned to improve the efficiency and presentation. | |
| Workbench v2 update | Import real transaction data csv file and add plan.<br>Add valuation models to further improve the project. | |

## 4.3 AI Prompt history (Claude, Phase II)

The prompts for October 13 to 18 are the author's own wording, translated from Chinese and shortened. The prompts for October 1 to 12 are summarised from the work log. The complete chat history is retained with the project documentation.

| Date (2026) | Prompt / Request | Decision / Outcome |
|---|---|---|
| Oct 1 (work log) | Review the original v1 repository and list improvements for analysis logic, token and prompt efficiency, pagination, timeline and UI. | Decision to redesign as v2 rather than patch v1. |
| Oct 2–3 (work log) | Better prompt and efficient token use; the analysis report is cut off and errors. | Thesis tracker design: each review a self-contained state, deterministic change detection, no model call when nothing changed. Fixed truncated output and OpenBB start-up. |
| Oct 4–7 (work log) | Multi-model switch when there is an error; design and test the personal portfolio part; add a journal; add Chinese translation. | Gemini fallback chain; broker CSV import, position plans, trade reasons, journal, on-demand Chinese. |

# Mid-Course Progress Report — Menghua Wang

| Date (2026) | Prompt / Request | Decision / Outcome |
|---|---|---|
| Oct 13 | "Add ? help to the Compare inputs so people know what they are doing. Look at the DEO analysis: this is an expectations-turnaround case, how should the analysis show it? Wouldn't analyst expectations plus later growth be a more reasonable default? Add a README for GitHub. Continue with how valuation results link to the company page. What number does the model actually take? Think deeply, then continue. Is there anything to learn from ChatGPT's evaluation?" | Analyst year as the default start with the exit multiple on the same basis; "what you must believe" grid; Valuation tab, out-of-date flags and assumption links on the company page; README; ChatGPT's review adopted in part (see the cross-check table). |
| Oct 13 | "What does Save snapshot mean? Explain the valuation logic, how each model thinks, what conclusions it supports and how reliable it is, using WMT." | Explanation written; saved-snapshot list relabelled; conclusion highlighted. |
| Oct 14 | "The Add-as-a-question button is not in the Valuation tab. Too many valuation versions: save only on confirmation and allow deleting one. Add Chinese content to the valuation pages. How should I read the Compare table? Review the whole app and finish it for GitHub." | Question card on every company; unused imports replaced, records deletable; Chinese layer; stale tests fixed; README rewritten (v2.24.0). |
| Oct 16 | "How can the Fundamentals tab be laid out better? CNOOC shows 'currencies must be identical': can non-USD companies be valued, for example in HKD? How should the 30-day digest titles be shown, and can holdings be sorted by P&L?" | Currency conversion at a labelled spot rate; Fundamentals regrouped; digest titles aligned and sortable by P&L (v2.25.0). |
| Oct 17 | "How should Barron's and Seeking Alpha be brought in, and should they be?", then ChatGPT's opinion and Claude's reply compared. | Headlines and links only, never full text and never sent to the model; then dropped as not worth the extra step. |
| Oct 17 | "Add forecasts beside the historical FCF in Fundamentals; make the valuation page return to the same company.", then "the forecast columns leave either no history or no forecast, cancel it; keep the return link; cancel the readings feature; the design work ends here." | Forecast columns built, judged unreadable and removed; return link kept (v2.27.0); design frozen. |
| Oct 18 | "Here is Muse's review of the code: is there room to improve?" and the pytest result. | Seven points checked against the code (3 valid); one stale test corrected. |